As previously mentioned, you could use these alerting solutions also from Grafana (which has a CloudWatch data source available if you want to control alerts from that pane), providing a single pane of glass. We're switching gears now for a quick look at end user tracking as well as resource usage tracking.

7.3 Usage tracking

Alerting is by and large a reactive process. This means something is happening, and based on this, someone gets notified, reacting to the event. In this section, we cover a different cloud operations technique that you can use for a variety of cases, from troubleshooting to security to planning. I'm talking about tracking users and what they access. Let's start with end users of your cloud-native app.

7.3.1 Users

To understand how your end users use your app, you need to have a way to track their activity within your app. This allows you to identify popular features and potential UX issues. Over the years, the term *real user monitoring* (RUM; https://github.com/open-telemetry/oteps/issues/169) has been popularized for this. For example, there is, for web applications, the widely used Google Analytics (https://analytics.google.com/analytics/web/), and you can use CloudWatch RUM (http://mng.bz/9Drx) in AWS. All these offerings require the up-front embedding of some small JavaScript snippet in your web pages to see the usage information (demographics, region, device types, usage paths, etc.) in a dedicated place. If you'd like to dive deeper into the topic, I recommend perusing the excellent article, "An Introduction to Real User Monitoring (RUM): Monitoring and Observability," by Chinmay Gaikwad (http://mng.bz/jPJz).

> **NOTE** One thing to be aware of with RUM is the rise of single-page application (SPA; https://en.wikipedia.org/wiki/Single-page_application) websites. In the context of SPAs, the traditional metrics, such as page load time, no longer matter (or make sense), since the page only loads once, at the beginning of a session, and the UI updates are effectively local renderings. For more on this topic, check out episode 60 of the O11ycast, "Customer-Centric Observability With Todd Gardner and Winston Hearn" (http://mng.bz/Wz94), as well as the article, "Observable Frontends: the State of OpenTelemetry in the Browser," by Jessica Kerr (http://mng.bz/8rdZ).

RUM is also something the CNCF OpenTelemetry project considers; the OpenTelemetry Enhancement Proposal OTEP 169 (http://mng.bz/EQoo) covers this by proposing a new data model and semantics, supporting the collection and export of RUM telemetry, a new signal type, alongside traces, metrics, logs, and profiles. While still in its early days, you can expect more development in this space, especially since the logs signal type stabilized in June 2023 (meaning you can use OpenTelemetry to handle logs in prod).